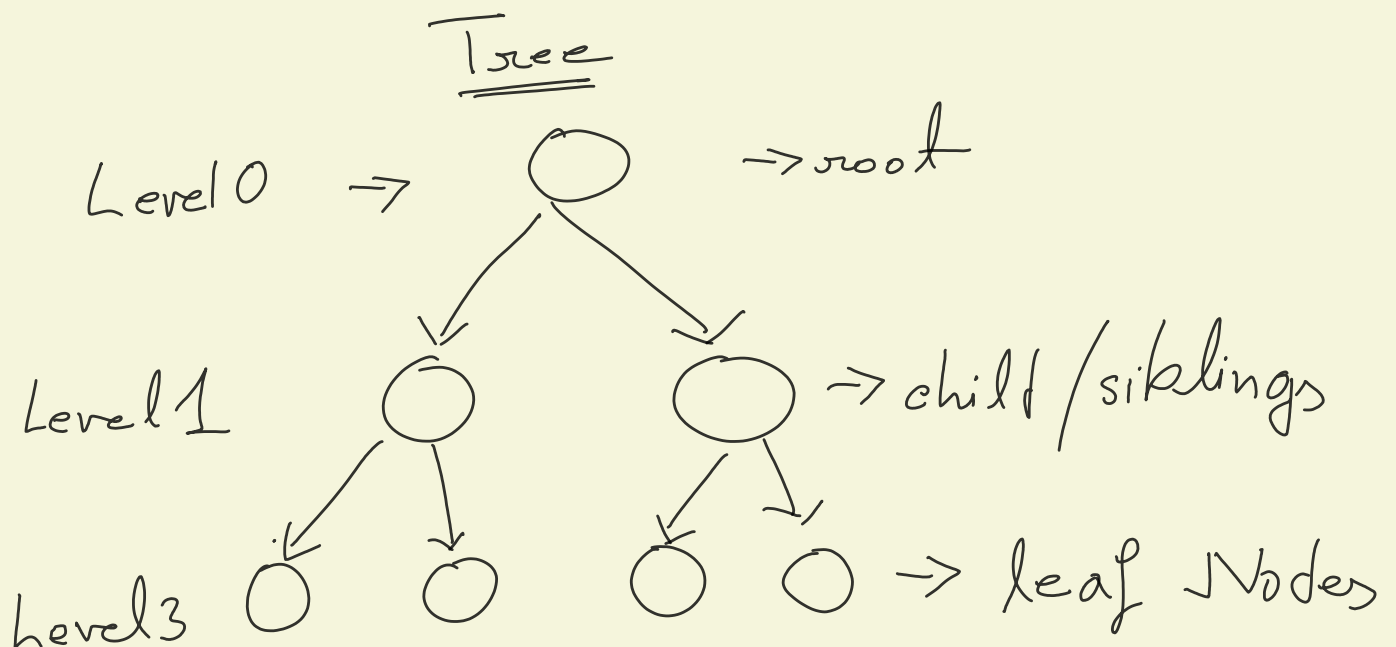


Linear Data Structure

- Array
- LL
- Stack
- Queue

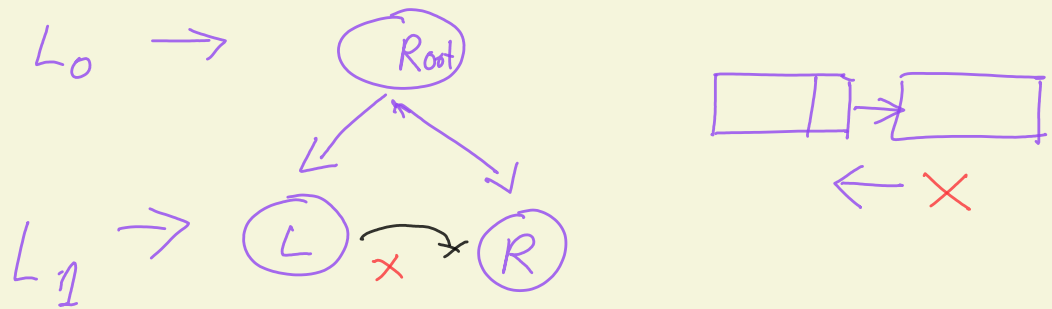
Non Linear Data Structure

- Tree
- Graph



Tree :- $\rightarrow$ Top to Down ✓

$\rightarrow$ Bottom up not allowed ✓



$\rightarrow$ BackTrack is only allowed by

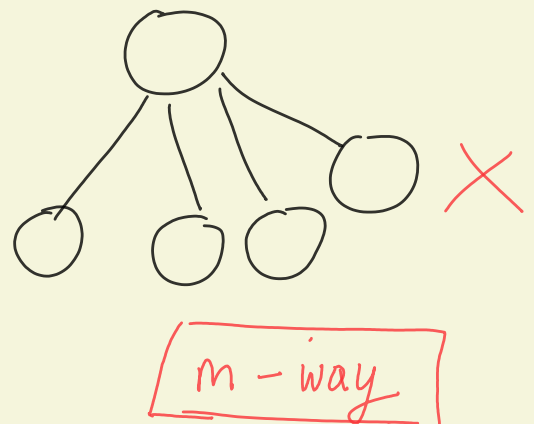
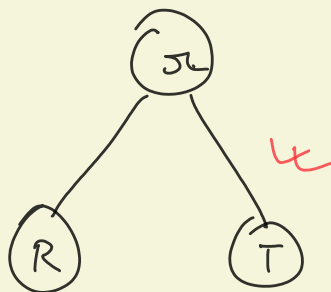
recursion or stack

$\rightarrow$ siblings can not talk directly to each other

Types of Tree

① Binary Tree :-

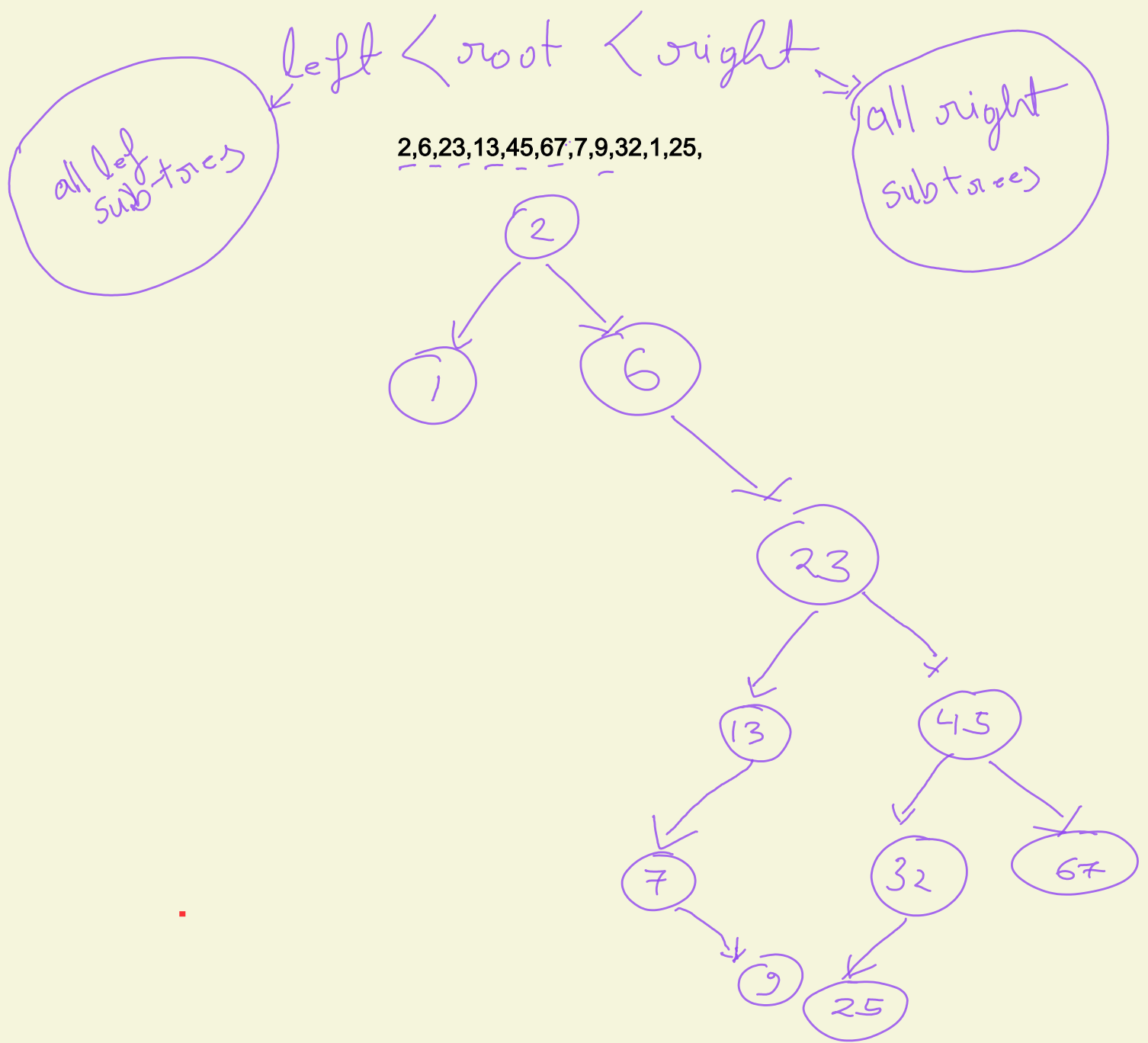
$\rightarrow$ max two childs



② Binary Search Tree (BST) :-

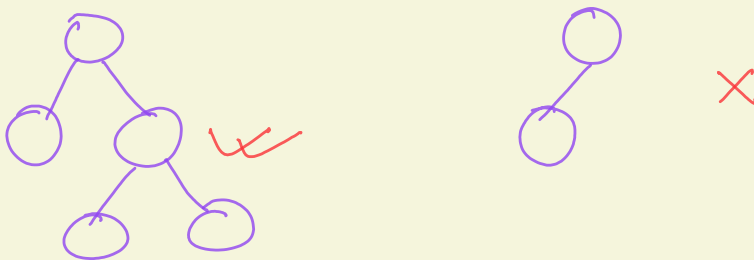
$\rightarrow$ It is a Binary Tree

$\rightarrow$ The nodes sub trees should be



③ Strict Binary Tree / Full Binary Tree :-

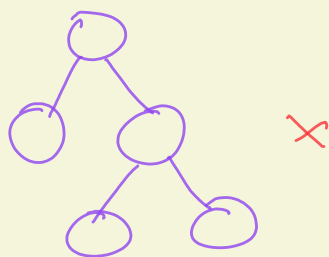
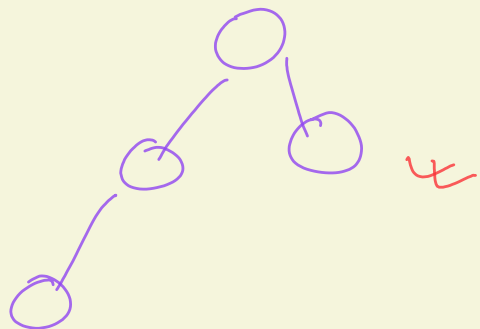
→ either no child or two child, but can't have single child.



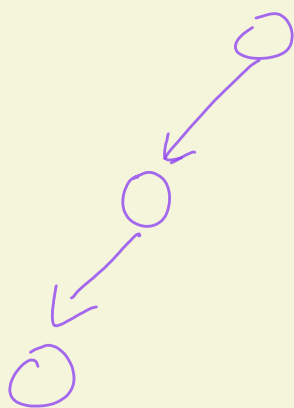
two child but can't

④ Complete Binary Tree:-

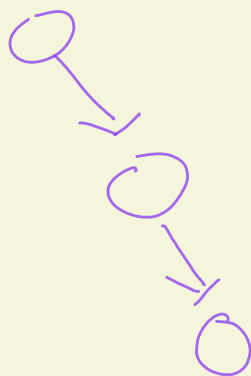
- preference is from left to right
- parent level should be full



⑤ Skew Binary Tree:- It's more like LL



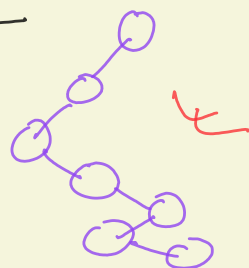
Left Skew



Right Skew

⑥ Degenerated Tree:-

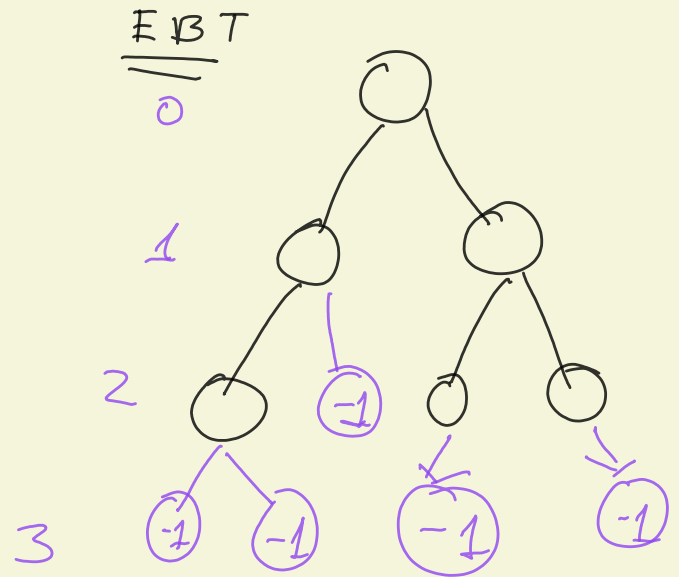
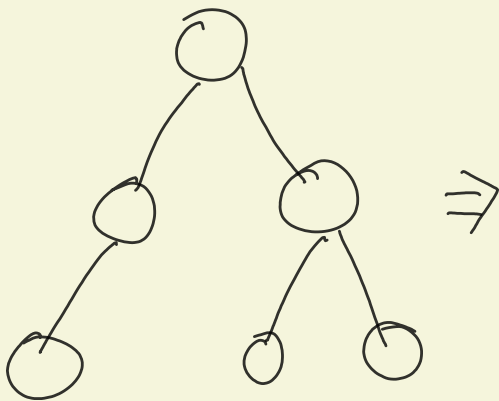
- every level should have one node
- don't have any direction



⑦ Extended Binary Tree⁰-

→ make a Full/Strict Binary Tree by adding dummy nodes.

→ all the leaf nodes should be dummy node.



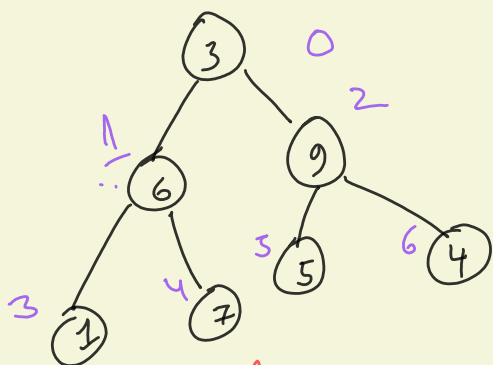
Tree representation

↳ array

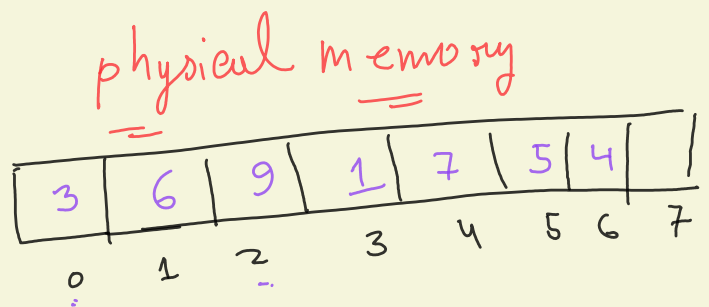
↳ LL

array Representation

ex 1



Logical



$$\text{left} = (\text{index} * 2) + 1$$

$$\text{right} = (\text{index} * 2) + 2$$

```

graph TD
    1((1)) --- 2((2))
    1 --- 3((3))
    2 --- 3_2((3))
    2 --- 4_2((4))
    3 --- 4_3((4))
    3 --- 5((5))
    4_2 --- 6((6))
    7((7))
    8((8))
    9((9))
    10((10))
    11((11))
    12((12))
  
```

1	2	3	X	X	4	5	X
0	1	2	3	4	5	6	7

LL Representation

